

Assessment of high-dose acetylcysteine in acute high-risk paracetamol (acetaminophen) ingestion.

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Prompt acetylcysteine treatment with standard doses (300 mg/kg over 21 h in divided doses) is almost universally effective in preventing hepatotoxicity after paracetamol (acetaminophen) overdose. However, hepatotoxicity is reported despite early treatment when paracetamol concentrations exceed 300 mg/L (1,985 μ mol/L) at 4 h. Prior studies evaluating high-dose acetylcysteine to treat high-risk ingestions have shown mixed results. We compared outcomes in patients with high-risk ingestions receiving standard or high-dose acetylcysteine. Records from a single poison center were reviewed from 1 January 2017 to 31 December 2022. We included cases of acute paracetamol ingestion treated with intravenous acetylcysteine with an initial paracetamol concentration above the "300 mg/L" (1,985 μ mol/L) line on the Rumack-Matthew nomogram. We compared standard and high-dose acetylcysteine groups by odds ratios and multivariable logistic regression. We defined hepatotoxicity as aminotransferase activity >1,000 U/L. We included 190 cases. Fifty-six percent received standard-dose acetylcysteine while 44% received high-dose acetylcysteine. Treatment within 8 h yielded no difference in hepatotoxicity between groups (odds ratio 1.67, 95% CI 0.067-42.3). Among patients treated after 8 h, hepatotoxicity was more common in the high-dose group (odds ratio 3.39, 95% CI 1.25-9.2) though odds of liver failure were similar (odds ratio 2.78, 95% CI 0.89-8.69). Eighty-eight percent of patients with hepatotoxicity had elevated aminotransferase activity at presentation. No patient died or received a liver transplant. Rates of hepatotoxicity were low in patients treated within 8 h regardless of acetylcysteine dose. Unexpectedly, high-dose acetylcysteine treatment was associated with an increased odds of hepatotoxicity in those treated after 8 h, but most had abnormal aminotransferase activities at presentation and there was no difference in rates of liver failure. Limitations include the use of retrospective, voluntarily reported poison center data. Prompt treatment with acetylcysteine, regardless of dose, prevented hepatotoxicity in high-risk paracetamol ingestion.

A Clinical Course of Repeated Supratherapeutic Ingestion of Acetaminophen.

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Acute liver failure (ALF) exemplifies a rapid decline in liver function among individuals with previously healthy livers, often manifesting through symptoms such as jaundice, confusion, and potentially life-threatening complications. Timely medical intervention, and, in severe instances, liver transplantation, are essential for enhancing outcomes and averting further deterioration. While the causes of ALF are multifaceted, in developed nations, it predominantly arises from drug-induced liver injury. Treatment primarily revolves around supportive measures, with severe cases necessitating liver transplantation. In instances where acute overdose with acetaminophen serves as the instigating factor, N-acetylcysteine (NAC) emerges as a pivotal component of management, as indicated by the Rumack-Matthew nomogram. The Rumack-Matthew nomogram guides treatment for acetaminophen overdose by correlating serum levels with the risk of liver damage. If levels exceed a set threshold, NAC is administered to prevent toxicity by replenishing glutathione. The decision to administer NAC is typically guided by this clinical tool, which aids healthcare providers in determining the appropriate course of action. NAC assumes a critical role in ameliorating the detrimental effects of acetaminophen overdose, particularly in averting liver damage, thus holding significant importance in patient care and recovery. While chronic acetaminophen overdose cases leading to ALF may also benefit from NAC, the supporting evidence remains weak. In this context, we present a case of ALF stemming from chronic

acetaminophen ingestion, managed with NAC when liver transplantation was not a viable option.

Assessing the frequency and types of errors involved in the use of a modified intravenous N-acetylcysteine protocol for acetaminophen overdose.

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Acetaminophen overdose is a leading cause of acute liver failure in developing countries. N-acetylcysteine (NAC) is a highly effective antidote for acetaminophen hepatotoxicity, typically initiated in the emergency department. Due to a known high rate of errors with the standard three-bag IV NAC protocol, in 2019, the Ontario Poison Center changed to a modified 3% IV NAC one-bag protocol. This study was undertaken to determine the frequency and types of errors associated with the use of this protocol. Data were gathered via chart review of Ontario Poison Centre electronic medical record cases identified as receiving IV NAC for acetaminophen overdose between August 1 and September 30, 2022. 218 total charts were identified, and 188 were deemed eligible based on inclusion and exclusion criteria. Errors were identified in 25% of charts, consisting of dosing errors in 11.7%, stopping errors in 9.0%, initiation errors in 3.7%, and interruptions in therapy in 3.2%. Dosing errors were the most common type of error (44.4%), with overdoses occurring three times more than underdoses. Errors were identified at 39% of geographic locations in the charts reviewed, with similar frequency in Ontario, Manitoba, and Nunavut. Clinical outcomes were similar in charts with and without errors. The rate of errors identified with this 3% IV NAC one-bag protocol is lower than reported for the standard three-bag protocol, but remains high due to dosing errors. Previously reported issues with prolonged interruptions in therapy with the standard three-bag protocol were low with the current 3% one-bag protocol. Although severe outcomes are rare, IV NAC overdose can be fatal. Identifying local factors in emergency departments that can contribute to administration errors (i.e., dose calculation, pump programming issues) can enhance the safety of this important antidote.

Intravenous N-acetylcysteine: the treatment of choice for paracetamol poisoning.

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One hundred cases of severe paracetamol poisoning were treated with intravenous N-acetylcysteine (acetyl-cysteine). There was virtually complete protection against liver damage in 40 patients treated within eight hours after ingestion (mean maximum serum alanine transaminase activity 27 IU/l). Only one out of 62 patients treated within 10 hours developed severe liver damage compared with 33 out of 57 patients (58%) studied retrospectively who received supportive treatment alone. Early treatment and acetylcysteine also prevented renal impairment and death. The critical ingestion-treatment interval for complete protection against severe liver damage was eight hours. Efficacy diminished progressively thereafter, and treatment after 15 hours was completely ineffective. Intravenous acetylcysteine was more effective than cysteamine and methionine and noticeably free of adverse effects. It is the treatment of choice for paracetamol poisoning.

[Intravenous or oral N acetylcysteine therapy in paracetamol poisoned patients. Should

treatment guidelines be reviewed?].

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Danish paracetamol (PCM) poisoned patients are treated with N-acetylcysteine (NAC) intravenously for 36 hours. This probably leads to overtreatment. Today, patients with poor prognoses can be identified and, in addition, NAC may have serious side effects. We reviewed the literature (route of administration, duration and timing of treatment) and found that intravenous NAC often leads to side effects (some serious), primarily when serum paracetamol is low. These patients are often only mildly poisoned and they may therefore benefit from a shorter, orally administered regimen (equally efficient and with fewer side effects than intravenously administered NAC).

Compromised blood-bile barrier after acetaminophen overdose.

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N-acetylcysteine (NAC) is the only approved drug for the treatment of acetaminophen (APAP) intoxication. A limitation of NAC is the short therapeutic time-window as it is only effective within approximately eight hours after APAP ingestion, which is critical since patients seek medical attention often after the onset of symptoms approximately 24 h after overdose. Recently, a so far unknown mechanism was identified by which APAP causes an increase of intracellular bile acid concentrations in hepatocytes to concentrations that exceed cytotoxic thresholds. APAP compromises the tight junctions of bile canaliculi that leads to the leakage of highly concentrated bile acids into the sinusoids. From the sinusoidal blood, a high fraction of the released bile acids is transported back into hepatocytes by basolateral uptake carriers and secreted into bile canaliculi. Repeated leakage from the canaliculi followed by hepatocellular reuptake and canalicular secretion causes an increase of intracellular bile acid concentrations and finally hepatocyte death. Importantly, inhibition of bile acid uptake carriers reduced intracellular bile acid concentrations and strongly ameliorated APAP hepatotoxicity, a finding that could result in a new therapeutic option for APAP-intoxicated patients.

Acetaminophen overdose.

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N-acetylcysteine (NAC) is the treatment of choice for acetaminophen overdose. With this therapy, morbidity from overdose can be held to a minimum. Mortality is rare in any case and virtually nonexistent in treated patients. Unless a high index of suspicion is maintained, the diagnosis may be missed until it is too late for effective antidotal treatment.

Paracetamol overdose in the newborn and infant: a life-threatening event.

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Paracetamol is the only drug recommended to treat fever in neonates. At recommended doses, paracetamol has not been associated with liver injury in neonates, while hepatotoxicity may occur after

intake of a single high dose or multiple excessive doses. The aim of this narrative review is to critically analyze and summarize the available literature on newborns and infants exposed to supratherapeutic doses of paracetamol, with special focus on their clinical features, outcome, and management. The PubMed, SCOPUS, and Google Scholar search engines were used to collect data, without time limitation. The following keywords were used: paracetamol/acetaminophen, overdose, hepatotoxicity, N-acetylcysteine, newborn, infant. The literature search identified a total of 27 case reports, a number of review articles, and few other relevant publications. Neonatal poisoning from paracetamol resulted from transplacental drug transfer after maternal overdose in some published cases, while it was the consequence of medication errors in other cases. Newborns and infants who have received a single overdose and have paracetamol concentrations below the Rumack-Matthew nomogram limits are at low risk of serious hepatic damage, while those who have recently ingested more than one supratherapeutic dose of paracetamol should be managed with caution. The treatment of choice for paracetamol poisoning is N-acetylcysteine, a specific antidote which reduces paracetamol hepatotoxic effects. N-Acetylcysteine should be given according to specific regimens through weight-based dosing tables. Caution should be used when paracetamol is administered to the newborn. In the event of an overdose, careful patient monitoring and personalization of post-overdose procedures are recommended.

Acetaminophen overdose associated with double serum concentration peaks.

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Acetaminophen is the most commonly used analgesic-antipyretic medication in the United States. Acetaminophen overdose, a frequent cause of drug toxicity, has been recognized as the leading cause of fatal and non-fatal hepatic necrosis. N-Acetylcysteine is the recommended antidote for acetaminophen poisoning. Despite evidence on the efficacy of N-acetylcysteine for prevention of hepatic injury, controversy persists about the optimal duration of the therapy. Here, we describe the case of a 65-year-old male with acetaminophen overdose and opioid co-ingestion who developed a second peak in acetaminophen serum levels after completing the recommended 21-hour intravenous N-acetylcysteine protocol and when the standard criteria for monitoring drug levels was achieved. Prolongation of N-acetylcysteine infusion beyond the standard protocol, despite a significant gap in treatment, was critical for successful avoidance of hepatotoxicity. Delay in acetaminophen absorption may be associated with a second peak in serum concentration following an initial declining trend, especially in cases of concomitant ingestion of opioids. In patients with acetaminophen toxicity who co-ingest other medications that may potentially delay gastric emptying or in those with risk factors for delayed absorption of acetaminophen, we recommend close monitoring of aminotransferase enzyme levels, as well as trending acetaminophen concentrations until undetectable before discontinuing the antidote therapy.

Assessment and treatment of acetaminophen overdose.

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The assessment and treatment of acute acetaminophen toxicity are reviewed, and several unresolved issues are discussed. The drug most commonly involved in analgesic ingestions, as a single agent or in combination, is acetaminophen. Hepatotoxicity, the major manifestation of acetaminophen ingestion, is believed to result from the accumulation of a toxic metabolite, N-acetylimidoquinone. The clinical

features of acetaminophen toxicity are divided into four stages ranging from minor symptoms, such as gastrointestinal irritation, to coma and death. The most reliable method for assessing the severity of acetaminophen ingestion is comparison of serum acetaminophen concentrations with the Rumack-Matthew nomogram. Delaying the start of antidotal therapy for more than 10 hours after the ingestion increases the risk of toxicity; no benefit is observed when antidotal therapy is initiated more than 24 hours after ingestion. Acetaminophen is often marked in combination with other drugs; a detailed patient history and a urine toxicology screen help determine whether multiple toxic ingestions have occurred. The roles of ipecac, gastric lavage, and activated charcoal in therapy are controversial. The mainstay of treatment for acetaminophen intoxication is a 17-dose course of acetylcysteine.
